

MAT1375, Classwork1, Fall2026

Ch1. Intervals and Functions

1. Set Notation of a set:

(1) Let V be a set including all the five main vowels in English alphabet. $V = \{a, e, i, o, u\}$.

(2) Let O be a set with all the odd positive numbers. $O = \{1, 3, 5, 7, \dots\}$.
 $= \{x \mid \text{all odd positive numbers}\}$

In general, we have $S = \{x \mid \text{either list all the elements or give a description}\}$.
 $= \{x \mid x = 2n+1, n \text{ is all positive numbers}\}$

2. Number systems:

$\mathbb{N} = \{0, 1, 2, 3, 4, 5, \dots\}$; Natural Numbers

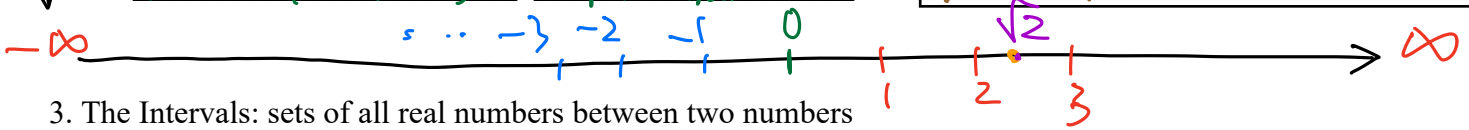
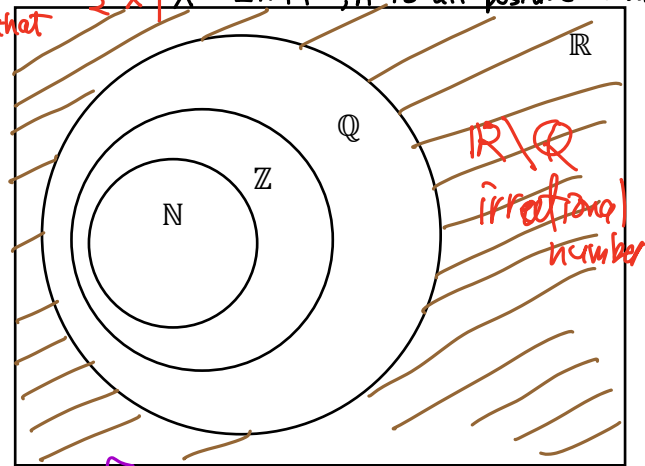
$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$; Integers

$\mathbb{Q} = \{x = \frac{a}{b} \mid a \in \mathbb{Z}, b \in \mathbb{Z}^+\}$; Rational Numbers

$\mathbb{R} \setminus \mathbb{Q}$: Irrational number. π (Pi), $\sqrt{2}$, e (2.715...)

$\mathbb{R}$ = Real Numbers

$\mathbb{C} = \{x = a + ib \mid a \in \mathbb{R}, b \in \mathbb{R}\}$; Complex Number



3. The Intervals: sets of all real numbers between two numbers

Let a and b be real numbers with $a \leq b$. We have

$[a, b] = \{x \mid a \leq x \leq b\}$	
$(a, b) = \{x \mid a < x < b\}$	
$[a, b) = \{x \mid a \leq x < b\}$	
$(a, b] = \{x \mid a < x \leq b\}$	
$(-\infty, b) = \{x \mid -\infty < x < b\}$	
$(-\infty, b] = \{x \mid -\infty < x \leq b\}$	
$[a, \infty) = \{x \mid a \leq x < \infty\}$	
$(a, \infty) = \{x \mid a < x < \infty\}$	

4. Complete the table

Inequality notation	Number line	Interval notation
$-3 < x < 5$		$(-3, 5)$
$-1 < x \leq 2$		$(-1, 2]$
$-\infty < x < 3$		$(-\infty, 3)$

or just " $x < 3$ "

5. The definition of a Function:

Let A and B be nonempty sets. A _____ f from A to B is an assignment of _____ element of B to each element of A , denoted by $f: \text{---} \rightarrow \text{---}$ which is read as f maps A to B . Then we say that A is the _____ of f and B is the _____ of f .

6. If $f(a) = b$, we say that b is the _____ of a and the _____ of f is the set of all images of elements of A .

7. Suppose that each student in 1375 is assigned a letter grade from the set $G = \{A, B, C, D, F\}$. And suppose that grades are A for Adams, C for Chou, B for Goodman, A for Rod, and F for Stevens.

Let $S = \{\text{Adams, Chou, Goodman, Rod, Stevens}\}$ be the set of the students.

(a) Is $f: S \rightarrow G$ a function?

(b) What is the domain, range, and codomain of f ?

(c) Is $g: G \rightarrow S$ a function?

8. The tables below describe assignments between inputs x and outputs y . Determine which of the given tables describe a function. If they do, determine their domain and range. Describe which outputs are assigned to which inputs.

a)

x	19	7	6	-2	3	-11
y	3	3	3	3	3	3

b)

x	1	2	3	4	4	5
y	5.3	9	13	17	15	$\sqrt{19}$